



CLINICAL PROGRAMMING BOOTCAMP • MODULE 17

Analysis of Lab Data

Ranges, shifts and criterion flags

The same BDS skeleton as ADVS — and the number that decides every percentage in it.

Hands-on: Notebook 17 (SAS) — Build ADLB

ADSL has 8 subjects. ADLB has 4.

Four subjects had no labs drawn. One subject has a post-baseline abnormal result. What percentage is that?

Denominator from ADSL

Drug A 1 / 4 = 25.0%

Placebo 0 / 4 = 0.0%

Correct. Every dosed subject was at risk, whether or not a sample was taken.

Denominator from ADLB

Drug A 1 / 3 = 33.3%

Placebo 0 / 1 = 0.0%

Wrong. And the Placebo denominator is ONE — a single abnormal there would read as 100%.

Denominators come from ADSL, filtered by the population flag the table calls for.

Never from the dataset you happen to be summarising. Missing data must not shrink a denominator — that silently converts “we did not measure this” into “this did not happen”.

One of the four missing subjects is ABC-01-01-004, who discontinued for an adverse event on day 20. Exactly the subject whose absence biases a safety result in the sponsor's favour.

Everything a normal range makes possible

ANRLO / ANRHI

the range itself, on every row

ANRIND

where THIS result sits: LOW · NORMAL · HIGH

BNRIND

where the BASELINE sat — carried to every row

SHIFT1

the two together: “NORMAL to HIGH”

CRIT1 / CRIT1FL

one prespecified yes/no question

BNRIND is carried to EVERY row of the parameter, like BASE. A shift needs to know where it started — so both ends must sit on the same row.

We READ TRTEMFL. We DERIVE ANRIND. Why?

	TRTEMFL (Notebook 15)	ANRIND (today)
What it encodes	a study DECISION	pure ARITHMETIC
Inputs	dates + a rule for partial dates	three numbers on this row
Can it drift?	yes — the rule lives elsewhere	no — it is a comparison
So we	READ it, cross-check it	DERIVE it, reconcile it

Read a DECISION. Derive a COMPUTATION. Verify either way.

Recalculating a decision does not verify it — it forks it. The instinct to “just recompute it so I know it's right” is a programming instinct, and in clinical data it is wrong more often than right.

The whole clinical content of this study's lab data

24 post-baseline analysis records. One cell is off the diagonal.

	→ NORMAL	→ HIGH	→ LOW
NORMAL →	23	1	0
HIGH →	0	0	0
LOW →	0	0	0

ABC-01-01-003 · ALT 24 → 72 U/L (range 7-56) · Drug A

A liver enzyme crossing the upper limit on active treatment — exactly the signal a hepatotoxicity review exists to find.

The diagonal means “stayed in the same category”. Off-diagonal means the result **CROSSED A CLINICAL BOUNDARY** — the point at which a number stops being reassuring and starts requiring explanation.

Blank is not 'N'

THE CRITERION

```
if paramcd = 'ALT' then do;  
  crit1 = 'ALT > ULN';  
  if aval > anrhi then crit1fl = 'Y'; else crit1fl = 'N';  
end;  
  
/* every other parameter: BLANK */
```

Correct — blank off-parameter

1 of 8 evaluated = 12.5%

Only ALT records were evaluated.

Wrong — 'N' everywhere

1 of 48 evaluated = 2.1%

The rate collapses by a factor of six.

'N' on a creatinine record asserts something false:

“This record was evaluated against criterion 1 — ALT > ULN — and did not meet it.”

ADaM needs THREE states — met (Y), evaluated and not met (N), not evaluated (blank). Collapsing the third into the second destroys it permanently: nothing distinguishes a real N from a manufactured one.

WHAT'S NEXT

48 rows, 4 subjects, one abnormal

1

Step 1 • Notebook 17 Build ADLB. The BDS skeleton from yesterday, plus ranges, shifts and the criterion flag.

2

Step 2 • Exercise 1 Compute the same percentage with both denominators. 25.0% and 33.3% — one is defensible.

3

Step 3 • Exercise 5 Write the rule for when ADaM should re-derive an upstream flag and when it should read it.

Tomorrow: a dataset that keeps every subject — including the ones nothing happened to — and a flag that runs backwards.